



# Assignment (MCO1 Milestone 1): Basic Input/Output Operations

## INTRODUCTION

Welcome to Part 1 of your MCO1 Banking and Currency Exchange Application! This initial task focuses on mastering basic console input and output (I/O) mechanisms to lay the foundation for your application's user interface.

## OBJECTIVE

Implement a basic console interface using standard input/output functions in **C**, **Java**, **Kotlin**, and **R** that display the module interfaces/menus, retrieve input from the user, and print a feedback of input values.

NOTE: The different module interfaces just need to be displayed one after the other. This exercise deals with basic I/O only.

## INTERFACE REQUIREMENTS

### Main Menu

| REQ #    | DETAILS                                                                                          |
|----------|--------------------------------------------------------------------------------------------------|
| REQ-0001 | Provision to display <b>main navigation menu</b> to standard console output (refer to Figure 1). |
| REQ-0002 | Provision to prompt and capture user selection (1–6).                                            |
| REQ-0003 | Provision to output feedback confirming selected menu choice.                                    |



```
Select Transaction:
[1] Register Account Name
[2] Deposit Amount
[3] Withdraw Amount
[4] Currency Exchange
[5] Record Exchange Rates
[6] Show Interest Amount
```

```
Choice: 1
```

```
***
```

```
Choice = 1
```

Fig. 1. Main Menu

## Register Account Name

| REQ #    | DETAILS                                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------------------------------|
| REQ-0004 | Provision to display prompt for transaction <b>Register Account Name</b> to standard console output (refer to Figure 2). |
| REQ-0005 | Provision to prompt and capture account name.                                                                            |
| REQ-0006 | Provision to display the captured account name.                                                                          |

```
Register Account Name
Account Name: Dela Cruz, Juan

***
Account Name = Dela Cruz, Juan
```

Fig. 2. Register Account Name



## Deposit Amount

| REQ #    | DETAILS                                                                                                                                              |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-0007 | Provision to display prompt for transaction <b>Deposit Amount</b> to standard console output (refer to Figure 3).                                    |
| REQ-0008 | Provision to capture the account name and deposit amount.                                                                                            |
| REQ-0009 | Provision to display the following default values: <ul style="list-style-type: none"><li>• Current balance = 1000</li><li>• Currency = PHP</li></ul> |
| REQ-0010 | Provision to display the captured account name and deposit amount.                                                                                   |

```
Deposit Amount
Account Name: Dela Cruz, Juan
Current Balance: 1000.00
Currency: PHP

Deposit Amount: 500.00

***
Account Name = Dela Cruz, Juan
Deposit Amount = 500.00
```

Fig. 3. Deposit Amount



## Withdraw Amount

| REQ #    | DETAILS                                                                                                                                              |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-0011 | Provision to display prompt for transaction <b>Withdraw Amount</b> to standard console output (refer to Figure 4).                                   |
| REQ-0012 | Provision to capture the account name and withdraw amount.                                                                                           |
| REQ-0013 | Provision to display the following default values: <ul style="list-style-type: none"><li>• Current balance = 1000</li><li>• Currency = PHP</li></ul> |
| REQ-0014 | Provision to display the captured account name and withdraw amount.                                                                                  |

```
Withdraw Amount
Account Name: Dela Cruz, Juan
Current Balance: 1000.00
Currency: PHP

Withdraw Amount: 500.00

***
Account Name = Dela Cruz, Juan
Withdraw Amount = 500.00
```

Fig. 4. Withdraw Amount



## Record Exchange rate

| REQ #    | DETAILS                                                                                                       |
|----------|---------------------------------------------------------------------------------------------------------------|
| REQ-0015 | Provision to display the menu for <b>Record Exchange Rate</b> to standard console output (refer to Figure 5). |
| REQ-0016 | Provision to capture the selected foreign currency name and exchange rate.                                    |
| REQ-0017 | Provision to display the selected foreign currency name and recorded exchange rate.                           |

### Record Exchange Rate

```
[1] Philippine Peso (PHP)
[2] United States Dollar (USD)
[3] Japanese Yen (JPY)
[4] British Pound Sterling (GBP)
[5] Euro (EUR)
[6] Chinese Yuan Renminni (CNY)
```

```
Select Foreign Currency: [2]
Exchange Rate: 62.00
```

```
***
```

```
Select Foreign Currency = [2]
Exchange Rate = 62.00
```

Fig. 5. Record Exchange Rate



## Currency Exchange

| REQ #    | DETAILS                                                                                                                                                                                                                                             |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-0018 | Provision to capture the source amount for <b>Currency Exchange</b> (refer to Figure 6).                                                                                                                                                            |
| REQ-0019 | Provision to display the exchange amount of all currencies using the following exchange values: <ul style="list-style-type: none"><li>• USD = 62.00</li><li>• JPY = 0.40</li><li>• GBP = 84.00</li><li>• EUR = 72.00</li><li>• CNY = 9.00</li></ul> |
| REQ-0020 | Provision to display the default source currency (PHP) and the source amount.                                                                                                                                                                       |

Foreign Currency Exchange  
Source Amount (PHP): 1000.00

Exchanged Currency  
[1] Philippine Peso (PHP) = 1000.00  
[2] United States Dollar (USD) = 62000.00  
[3] Japanese Yen (JPY) = 400.00  
[4] British Pound Sterling (GBP) = 84000.00  
[5] Euro (EUR) = 72000.00  
[6] Chinese Yuan Renminni (CNY) = 9000.00

\*\*\*

Source Currency = Philippine Peso (PHP)  
Source Amount (PHP) = 1000.00

Fig. 6. Record Exchange Rate



DEPARTMENT OF  
**SOFTWARE TECHNOLOGY**

## GRADING (80 pts total)

Each requirement is worth **1 point per programming language** (e.g., fulfilling Requirement 1 across all 4 languages earns 4 points total). Earn up to 20 points per language.